



De La Salle – College of Saint Benilde
8 Sen. L. Sumulong Memorial Circle, Antipolo, 1870 Rizal
(+63) 8667-1334 / (+63) 8667 – 1300 / 09565576301
<https://www.benilde.edu.ph/antipolo>

DATA STRUCTURE AND ALGORITHMS

Name: Mandal, Ella Mae C.	Date: September 3, 2026
Section: BSIS02	Score:

1. Student Information

```
using System;
class Program
{
    static void Main()
    {
        Console.WriteLine("Enter your name: ");
        string name = Console.ReadLine();

        Console.WriteLine("Enter your course: ");
        string course = Console.ReadLine();

        Console.WriteLine("Enter your year level: ");
        int yearlevel = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter your section: ");
        string section = Console.ReadLine();

        Console.WriteLine("\n===== STUDENT INFORMATION =====");
        Console.WriteLine("NAME: " + name);
        Console.WriteLine("COURSE: " + course);
        Console.WriteLine("YEAR LEVEL: " + yearlevel);
        Console.WriteLine("SECTION: " + section);
        Console.WriteLine("=====");
    }
}
```

Enter your name: Ella Mae C. Mandal
Enter your course: BSIS
Enter your year level: 2
Enter your section: BSIS02

===== STUDENT INFORMATION =====
NAME: Ella Mae C. Mandal
COURSE: BSIS
YEAR LEVEL: 2
SECTION: BSIS02
=====

C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 15748) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .



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2. Basic Calculator

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Enter first number:");
        int num1 = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter second number:");
        int num2 = Convert.ToInt32(Console.ReadLine());

        double sum = num1 + num2;
        double difference = num1 - num2;
        double product = num1 * num2;
        double quotient = num1 / num2;

        Console.WriteLine("==== RESULTS ===");
        Console.WriteLine("Addition: " + (num1 + num2));
        Console.WriteLine("Subtraction: " + (num1 - num2));
        Console.WriteLine("Multiplication: " + (num1 * num2));
        Console.WriteLine("Division: " + (num1 / num2));
    }
}
```

Output:

```
Enter first number: 32
Enter second number: 20

==== RESULTS ====
Addition: 52
Subtraction: 12
Multiplication: 640
Division: 1.6
```

3. Area of a Rectangle

```
using System;

class Program
{
    static void Main()
    {
        Console.WriteLine("Enter length:");
        int length = Convert.ToInt32(Console.ReadLine());

        Console.WriteLine("Enter width:");
        int width = Convert.ToInt32(Console.ReadLine());

        double area = length * width;

        Console.WriteLine("==== Area of a Rectangle ===");
        Console.WriteLine("Area: " + (length * width));
    }
}
```

Output:

```
Enter length: 12
Enter width: 25

==== Area of a Rectangle ====
Area: 30
```



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DATA STRUCTURE AND ALGORITHMS

4. Student Grade

The screenshot displays the Visual Studio IDE with a C# program named 'MandalApp'. The code is as follows:

```
1 using System;
2
3 namespace
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter First Grade: ");
8         int Grade1 = Convert.ToInt32(Console.ReadLine());
9
10        Console.WriteLine("Enter Second Grade: ");
11        int Grade2 = Convert.ToInt32(Console.ReadLine());
12
13        Console.WriteLine("Enter Third Grade: ");
14        int Grade3 = Convert.ToInt32(Console.ReadLine());
15
16        double average = (Grade1 + Grade2 + Grade3) / 3;
17
18        Console.WriteLine("==== Student Grade ===");
19        Console.WriteLine("Average: " + ((Grade1 + Grade2 + Grade3) / 3));
20    }
21 }
```

The output window shows the program's execution:

```
Enter First Grade: 95
Enter Second Grade: 97
Enter Third Grade: 98
==== Student Grade ====
Average: 96
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 3540) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

5. Boolean Input

The screenshot displays the Visual Studio IDE with a C# program named 'MandalApp'. The code is as follows:

```
1 using System;
2
3 namespace
4 {
5     static void Main()
6     {
7         Console.WriteLine("Are you a student? (true/false): ");
8         bool IsStudent = Convert.ToBoolean(Console.ReadLine());
9
10        Console.WriteLine("Student Status: " + IsStudent);
11    }
12 }
```

The output window shows the program's execution:

```
Are you a student? (true/false): true
Student Status: True
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 19452) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```



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DATA STRUCTURE AND ALGORITHMS

6. Age Verification

The screenshot displays the Visual Studio IDE with a C# project named 'MandalApp'. The code in the 'Main' method is as follows:

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter your age: ");
8         int age = Convert.ToInt32(Console.ReadLine());
9
10        bool isAdult = age >= 18;
11
12        Console.WriteLine($"Is 18 or older: {isAdult}");
13    }
14 }
```

The output window shows the program's execution:

```
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
The program "[26108] MandalApp.exe" has exited with code 0 (0x0).
```

The debug console shows the user input and the program output:

```
Enter your age: 21
Is 18 or older: True
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\Mandal
App.exe (process 26108) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Optio
ns->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

7. Full Name

The screenshot displays the Visual Studio IDE with a C# project named 'MandalApp'. The code in the 'Main' method is as follows:

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter your first name: ");
8         string firstName = Console.ReadLine();
9
10        Console.WriteLine("Enter your last name: ");
11        string lastName = Console.ReadLine();
12
13        string fullName = firstName + " " + lastName;
14
15        Console.WriteLine($"Full Name: {fullName}");
16    }
17 }
```

The output window shows the program's execution:

```
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
*MandalApp.exe* (CoreCLR: clrhost): Loaded "C:\Program Files\dotnet\shared\Microsoft.NETCore.App\
The program "[14156] MandalApp.exe" has exited with code 0 (0x0).
```

The debug console shows the user input and the program output:

```
Enter your first name: Ella Mae
Enter your last name: Mandal
Full Name: Ella Mae Mandal
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\Mandal
App.exe (process 14156) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Optio
ns->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```



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DATA STRUCTURE AND ALGORITHMS

8. Shopping Calculator

The screenshot displays the Visual Studio IDE with a C# program named 'MandalApp'. The code in the 'Program.cs' file is as follows:

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter product name: ");
8         string productName = Console.ReadLine();
9
10        Console.WriteLine("Enter price: ");
11        double price = Convert.ToDouble(Console.ReadLine());
12
13        Console.WriteLine("Enter quantity: ");
14        int quantity = Convert.ToInt32(Console.ReadLine());
15
16        double total = price * quantity;
17
18        Console.WriteLine($"Product: {productName}");
19        Console.WriteLine($"Total Cost: {total}");
20    }
21 }
```

The Output window shows the following execution results:

```
Enter product name: Zebra Sarasa Clip Pen 0.5 Black
Enter price: 89.75
Enter quantity: 3
Product: Zebra Sarasa Clip Pen 0.5 Black
Total Cost: 269.25
```

The Debug Console shows the following output:

```
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 16720) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

9. Salary Calculator

The screenshot displays the Visual Studio IDE with a C# program named 'MandalApp'. The code in the 'Program.cs' file is as follows:

```
1 using System;
2
3 class Program
4 {
5     static void Main()
6     {
7         Console.WriteLine("Enter employee name: ");
8         string employeeName = Console.ReadLine();
9
10        Console.WriteLine("Enter hours worked: ");
11        double hoursWorked = Convert.ToDouble(Console.ReadLine());
12
13        Console.WriteLine("Enter hourly rate: ");
14        double hourlyRate = Convert.ToDouble(Console.ReadLine());
15
16        double salary = hoursWorked * hourlyRate;
17
18        Console.WriteLine($"Employee: {employeeName}");
19        Console.WriteLine($"Calculated Salary: {salary}");
20    }
21 }
```

The Output window shows the following execution results:

```
Enter employee name: Ella Mae Mandal
Enter hours worked: 400
Enter hourly rate: 625
Employee: Ella Mae Mandal
Calculated Salary: 250000
```

The Debug Console shows the following output:

```
C:\Csharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 12524) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```



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10. MINI STUDENT INFORMATION SYSTEM

```
Program.cs
1 string name = Console.ReadLine();
2 Console.WriteLine("Enter Course: ");
3 string course = Console.ReadLine();
4 Console.WriteLine("Enter Age: ");
5 int age = Convert.ToInt32(Console.ReadLine());
6 Console.WriteLine("Enter Year Level: ");
7 string yearLevel = Console.ReadLine();
8 Console.WriteLine("Enter Grade 1: ");
9 double grade1 = Convert.ToDouble(Console.ReadLine());
10 Console.WriteLine("Enter Grade 2: ");
11 double grade2 = Convert.ToDouble(Console.ReadLine());
12 Console.WriteLine("Enter Grade 3: ");
13 double grade3 = Convert.ToDouble(Console.ReadLine());
14 double averageGrade = (grade1 + grade2 + grade3) / 3.0;
15 bool isAdult = age >= 18;
16 Console.WriteLine("---- STUDENT INFORMATION SUMMARY ----");
17 Console.WriteLine($"Name: {name}");
18 Console.WriteLine($"Course: {course}");
19 Console.WriteLine($"Age: {age}");
20 Console.WriteLine($"Year Level: {yearLevel}");
21 Console.WriteLine($"Average Grade: {averageGrade:F2}");
22 Console.WriteLine($"18 or Older: {isAdult}");

Output
Enter Student Name: Ella Mae Mandal
Enter Course: BSIS
Enter Age: 20
Enter Year Level: 2
Enter Grade 1: 96
Enter Grade 2: 98
Enter Grade 3: 97
---- STUDENT INFORMATION SUMMARY ----
Name: Ella Mae Mandal
Course: BSIS
Age: 20
Year Level: 2
Average Grade: 97.00
18 or Older: True
C:\sharp_workspace\example_app\MandalApp\MandalApp\bin\Debug\net10.0\MandalApp.exe (process 20760) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```

11. Final Challenge: Simple Store Receipt

```
Program.cs
1 Console.WriteLine("Enter customer name: ");
2 string customerName = Console.ReadLine();
3 Console.WriteLine("----- Product 1 -----");
4 Console.WriteLine("Enter product name: ");
5 string item1 = Console.ReadLine();
6 Console.WriteLine("Enter price: ");
7 double price1 = Convert.ToDouble(Console.ReadLine());
8 Console.WriteLine("Enter quantity: ");
9 int quantity1 = Convert.ToInt32(Console.ReadLine());
10 Console.WriteLine("----- Product 2 -----");
11 Console.WriteLine("Enter product name: ");
12 string item2 = Console.ReadLine();
13 Console.WriteLine("Enter price: ");
14 double price2 = Convert.ToDouble(Console.ReadLine());
15 Console.WriteLine("Enter quantity: ");
16 int quantity2 = Convert.ToInt32(Console.ReadLine());
17 double subtotal1 = price1 * quantity1;
18 double subtotal2 = price2 * quantity2;
19 double total = subtotal1 + subtotal2;
20 bool isValidTransaction = total > 0;
21 Console.WriteLine("=====");
22 Console.WriteLine("STORE RECEIPT");
23 Console.WriteLine("=====");
24 Console.WriteLine($"Customer: {customerName}");
25 Console.WriteLine($"Valid Order: {isValidTransaction}");
26 Console.WriteLine("=====");

Output
----- Product 1 -----
Enter product name: Apples
Enter price: 100
Enter quantity: 2
----- Product 2 -----
Enter product name: Oranges
Enter price: 100
Enter quantity: 3
=====
STORE RECEIPT
=====
Customer: Ella Mae Mandal
Valid Order: True
=====
1. Apples
2 x $100.00 = $200.00
2. Oranges
3 x $100.00 = $300.00
=====
TOTAL DUE: $500.00
=====
C:\Users\celes\OneDrive\Documents\DS\ALGO\SimpleStoreReceipt\SimpleStoreReceipt\bin\Debug\net10.0\SimpleStoreReceipt.exe (process 11468) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
Press any key to close this window . . .
```